

MODELLING TASK

DSA 5

LIST OF TASKS

Task 1: Fine-tuning a Lightweight Backbone for Image Classification

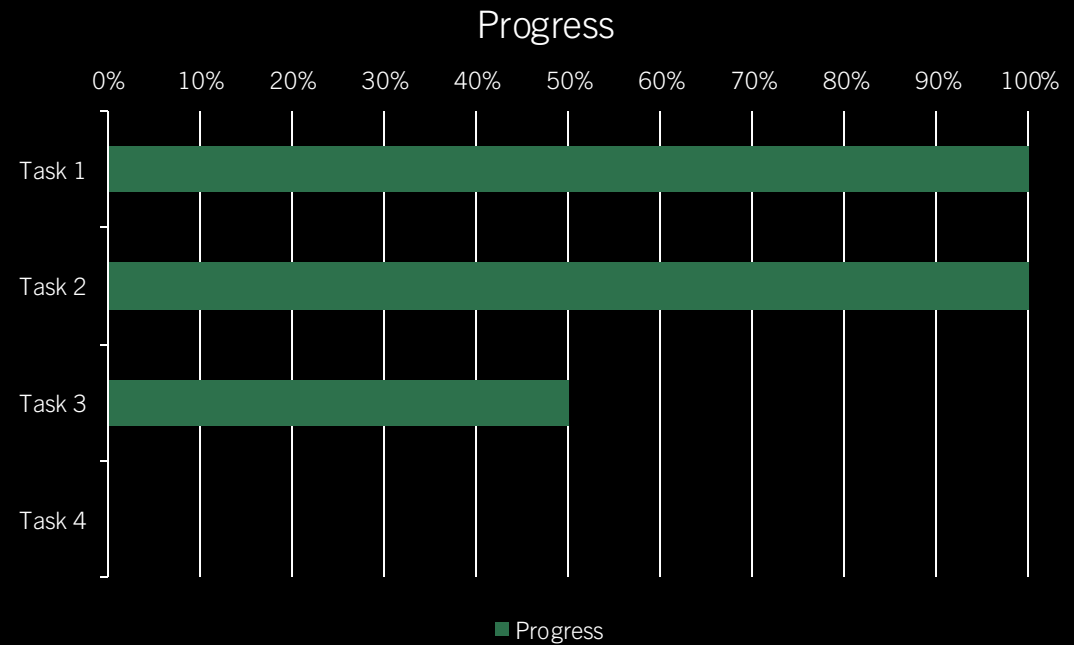
Task 2: Applying Quantization for Model Compression

Task 3: Building Custom CNN Backbone From Scratch

Task 4: Applying Quantization for Custom Model and Compare with Pretrain

PROGRESS

We start Task 1 and Task 3 first week. Task 2 started last week. Task 4 is not scheduled to start yet



TASK 1

Fine-tuning a Lightweight Backbone. Choosing MobileNet over ShuffleNet is an unpopular choice based on [Which Backbone to Use: A Resource-efficient Domain Specific Comparison for Computer Vision](#). However, MobileNetv3 is slightly more modern than ShuffleNet, and operate under lower FLOPs

Owner: Ngoc Lam Nguyen

Backbone: MobileNetV3-small

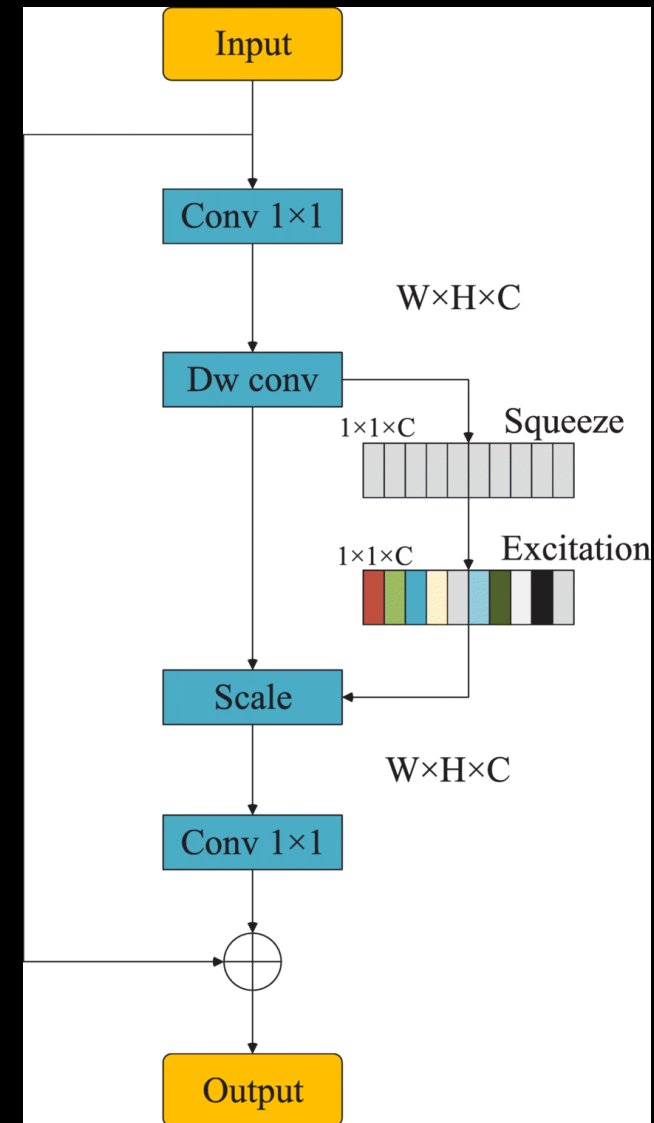
Dataset: CIFAR-10

Testing Metrics:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix
- Lightweight Metrics:
 - Number of parameters of the model
 - Estimate FLOPs (Floating Point Operations)

MOBILENETV3

(Li et al., 2022, Figure 4)



TASK 1 RESULT

Metrics	Score
accuracy	0.8102189781021898
precision	0.8121759435717278
recall	0.8102095904095904
f1	0.8106668804577737
trained_param_count	10250
flops	62874080

TASK 1 – CONFUSION MATRIX

- Biggest mistake is between cat-dog



TASK 2

Applied dynamic post-training quantization (PTQ) on pre-trained MobileNetV3 Small model fine-tuned on the CIFAR-10 dataset in task 1.

Quantization optimize neural networks by reducing the precision of weights and activations from 32-bit floating-point to 8-bit integers.

Owner: Jatinkumar Parmar

Backbone: Model from Task 1

Dataset: CIFAR-10

Testing Metrics:

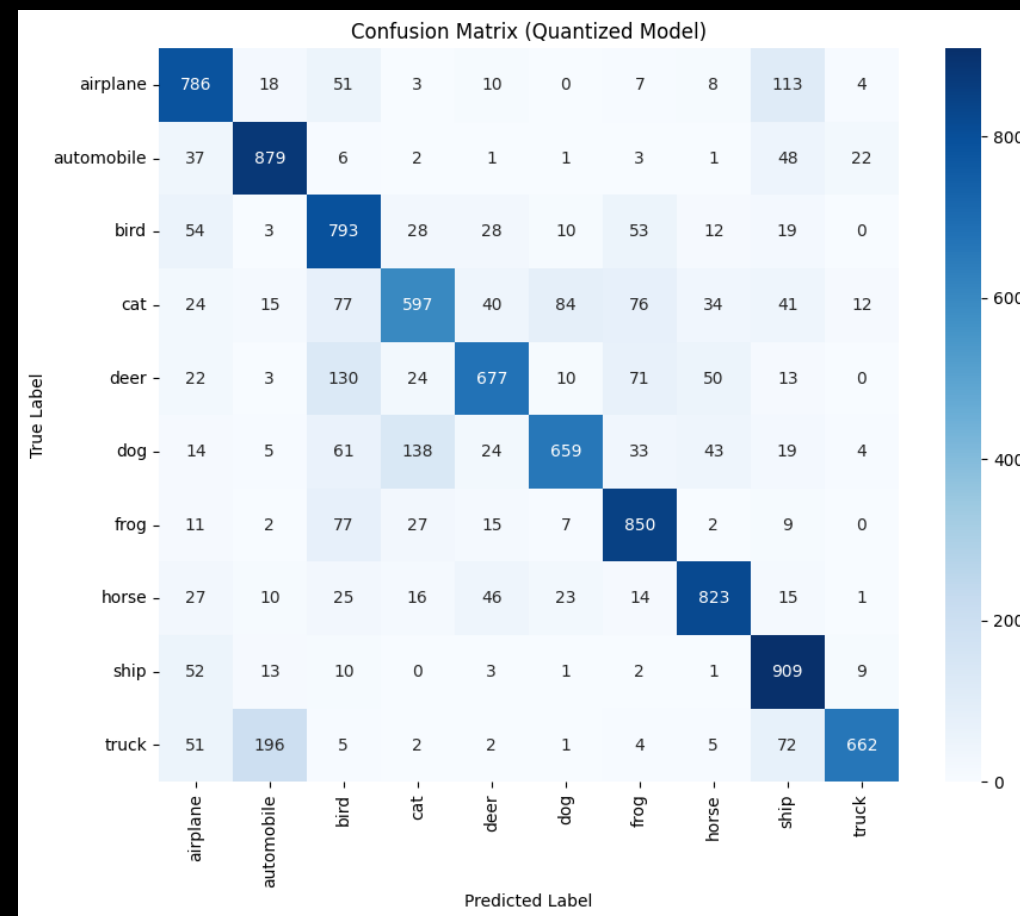
- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

TASK 2 RESULT

Metrics	Score
accuracy	0.7635
precision	0.7715
recall	0.7635
f1	0.7617
Model size	6234.402 KB
Model size (Quantized)	4435.79 KB

TASK 2 – CONFUSION MATRIX

- Biggest mistake is between cat-dog
- The mistake of a truck classified as automobile seems to be more serious than before



TASK 3

Creating a custom CNN architecture with more reduced trainable params roughly around 2500 performing on CIFAR-10 dataset

Fine-tuned layer by layers to shrink the model params with optimal loss in accuracy

Owner: Revanth Puvaneswaran

Backbone: Custom CNN

Dataset: CIFAR-10

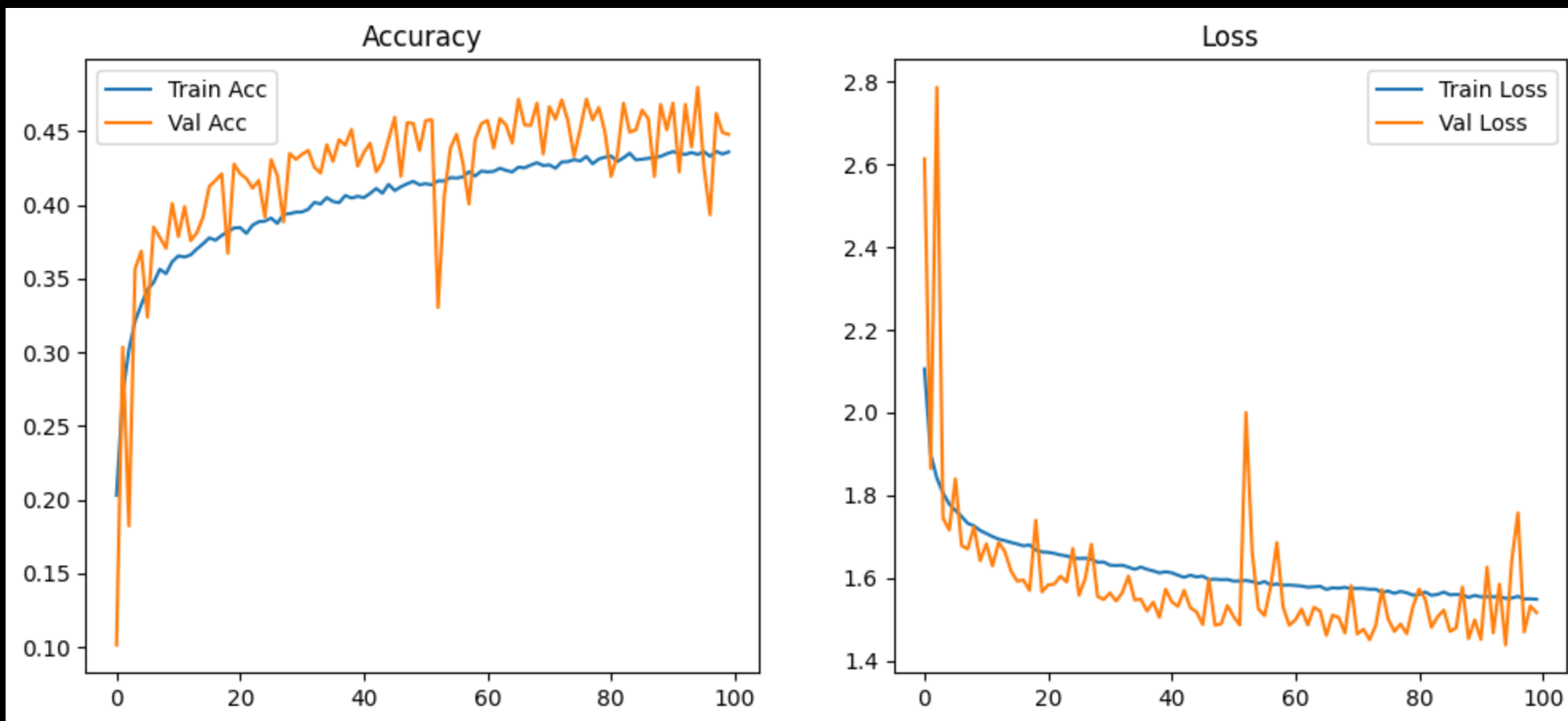
Testing Metrics:

- Accuracy
- Precision
- Recall
- F1-score

TASK 3 RESULT

Metrics	Score
accuracy	0.45
precision	0.4532
recall	0.4524
f1	0.4286
Model Params	2461
Model Size	112 KB

TASK 3 – PLOTS



TASK 3 – REPORT

```
Classification Report:
      precision    recall  f1-score   support

   airplane      0.67      0.32      0.43      1000
  automobile      0.52      0.67      0.59      1000
      bird      0.44      0.19      0.26      1000
       cat      0.24      0.38      0.30      1000
      deer      0.51      0.17      0.25      1000
       dog      0.35      0.55      0.43      1000
      frog      0.56      0.50      0.53      1000
     horse      0.49      0.44      0.47      1000
      ship      0.49      0.72      0.58      1000
     truck      0.47      0.53      0.50      1000

 accuracy          0.45      10000
  macro avg      0.48      0.45      0.43      10000
weighted avg      0.48      0.45      0.43      10000

Macro Precision: 0.4752
Macro Recall:    0.4458
Macro F1-score:  0.4333
```